



Anaesthesia

Workbook



Dr Swati



INSTRUCTIONS:

- Our initiative is to make a workbook in a precise manner to make you write the most important points, focus more on understanding the important concepts in class, save you precious time & energy, at the sametime keep you actively involved in the sessions.
- Workbook spaces are provided wherever necessary to help place your desired content.

PLEASE NOTE:

- This workbook has to be strictly used when the faculty is teaching as there are blank spaces which need to be filled, otherwise the inference may be misleading.
- The information contained in this book is strictly for educational purposes. Content is not intended as a substitute for professional medical advice, diagnosis or treatment.
- Every effort has been made to ensure the accuracy of information in this book. Neither the author nor the publisher are responsible for any errors or omissions. Any liability resulting from applying the ideas or information from this book, are strictly the responsibility of the reader.

DISCLAIMER:

- The notes have been made taking care of all the relevant points required but special care has been taken to avoid exhaustive nature. These notes are only for the purpose of understanding in a crisp manner and help you in quick complete revision. Please do not use them for the treatment of your patients as each patient requirement differ at various times.



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ANESTHESIA



Active space

PRE - ANESTHETIC EVALUATION**1. Focused history**

Previous anaesthesia exposure

Comorbidities in the patient

METs calculation

Surgery

Allergy & Medication history

General examination

Systemic examination

2. Routine investigations and depending upon the comorbidities and the type of surgery in the patient.

- a.
- b.
- c
- d
- e
- f

American society of anesthesiologist grading (ASA)

ASA grade I to VI: Elective surgery

ASA grade IE to VIE: Emergency surgery

✦ ASA Grading

ASA 1 - No comorbidity

ASA 2 - Controlled comorbidity

ASA 3- Comorbidity with moderate limitation

ASA 4- Comorbidity is a constant threat on life. These patients are never taken for elective surgery unless the benefit of the surgery is more than the risk of the surgery performed.

ASA 5- Surgery is the only option for survival

ASA 6- Comatose/Brain dead patients (for organ transplant)

AIRWAY EVALUATION

History of previous anaesthesia exposure and during that any airway mishap is evaluated Physical evaluation of airway

1. Mouth opening is assessed
2. Atlantooccipital joint movement
3. Neck joint movement
4. Mentohyoid/Mentothyroid distance: To get an idea whether the proper patient position for airway securement is possible or not

5. Mallampati Grading

Pre anesthetic evaluation of the airway

Predicts

Ask pt. to open the mouth

Evaluate the size of tongue in the oral cavity

If tongue size bigger, the structure won't be visible, which makes intubation difficult



MMP 1

MMP 2

MMP 3

MMP 4

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MIST

PREANESTHETIC ORDERS

Written informed consent regarding surgery and anaesthesia

Nil per oral prior to surgery

♦ **Adults**

8 hours: Full fatty meals

6 hours:

2 hours:

Pediatrics

8 hours: Full fatty meals

6 hours: Formula milk/semisolid foods

4 hours:

2 hours: Clear fluids

Orders Regarding Previous Medications

1. Oral Antihypertensives: Continue till the day of surgery

2. Oral Hypoglycaemic drugs

Minor / moderate surgery: Omit the last dose.

Major surgery: Omit the last dose and put the patient on insulin to maintain the blood glucose level

MIST

3. Antipsychotics, Antidepressants, Antiepileptics: Continue till the day of surgery, except Tricyclic antidepressants: Stop 3 weeks prior to surgery

Lithium: Stop hours prior to surgery

4. Thyroid medications: Continue till the

5. Anti-Coagulants

Aspirin: Continue till the day of surgery

Clopidogrel: Stop days prior to surgery

Ticlopidine: Stop days prior to surgery

LMWH: Stop hours prior to surgery

Warfarin: Stop days prior to surgery

Unfractionated Heparin: Stop 6 hours prior to surgery

6. Steroids:



MONITORING

ASA GUIDELINES

Standard I:

Standard II: Continuous monitoring of oxygenation, ventilation, metabolism and temperature of the patient

There Are Five Mandatory Monitors

1. ECG: Lead II and V5 are the most important leads
2. Non-Invasive Blood Pressure Monitoring (NIBP): Mandatory in every surgery
3. Pulse oximetry: SPO2 probe, measures oxygen saturation of arterial blood.
4. Capnography:
5. Temperature (core temperature)

✦ Desirable Monitors

Bispectral Index monitoring (BIS)

Neuromuscular monitoring:

Other monitors are attached when required: CVP monitor

PAP monitor

Transesophageal echocardiography (TEE)

CAPNOGRAPHY

Graphical representation of Vs Time

Capnogram : Shape of the graph

Capnometry : Maximum value of expired CO_2 -

Metabolism, Circulation, Ventilation : normal

D/Dx of flat capnograph -

1.

2.

3.

4.

5.

6.

❑ Normal graph

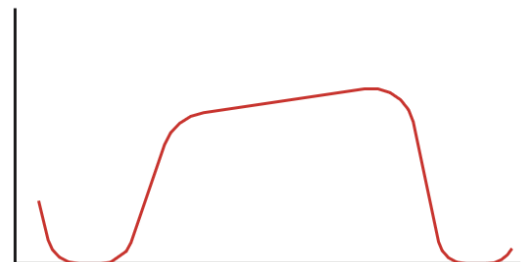
✦ There are four phases

➤ Phase I:

➤ Phase II:

➤ Phase III:

➤ Phase IV:



MIST

ETCO₂ -

Increased EtCO ₂	Decreased EtCO ₂
✦ Hypermetabolism (fever, malignant hyperthermia, shivering) ✦ Hy ✦ Poventilation	✦ Hypometabolism (hypothermia, anesthesia) ✦ Hyperventilation ✦ Hypoperfusion (embolism, shock)

Abnormal graphs and uses:

1. Confirmation of correct placement of ETT.

2. DURING CPR

3. DURING ANAESTHESIA

Active space



Active space

✦ Temperature monitoring



Core temperature is monitored

Sites for temperature monitoring

- ✦ Pulmonary artery catheter core temperature monitoring: Most sensitive site for temperature measurement, but not commonly used.

- ✦ Lower 1/3rd of esophagus: Most common site

- ✦ Tympanic membrane

- ◈ Pulse oximetry -

- ✦ Measures pulse rate and estimates the oxygen saturation of hemoglobin on a non-invasive continuous basis.



- ✦ Probe has a light emitting diode, which falls on the capillary and senses the flow of blood in capillary

- ✦ If its pulsatile, it is in arterial capillary: It will pass through.

- ✦ Pulse oximetry is based on 2 principles

1. Law of Plethysmography: To recognize, analyse and differentiate the pulsatile arterial signal from the non- pulsatile venous signal.

2. Law of Oximetry/ Spectrophotometry/ Beer Lambert's

◆ Inaccuracies in spo2 monitoring

- 1.
- 2.
- 3.
- 4.
- 5.

◆ BIS monitoring

Monitor the depth of anaesthesia.

◆ Three EEG electrodes placed in the forehead of the patient (i.e. Frontal, Parietal and Temporal lobe), and graphical representation is obtained.

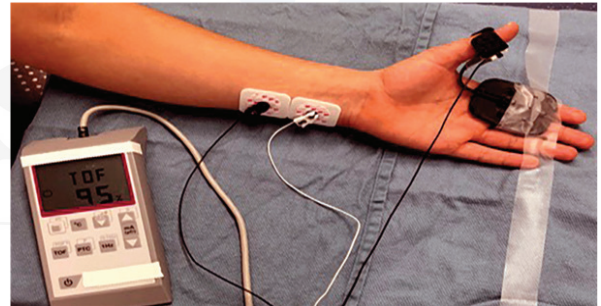


Active space

- ✦ Value ranges from-

Neuromuscular monitoring

- ✦ Measures adequacy of muscle paralysis.
- ✦ Most common nerve used: Ulnar nerve.
- ✦ Most common electrical stimulus used: Train of four stimulus.



CVP Monitor

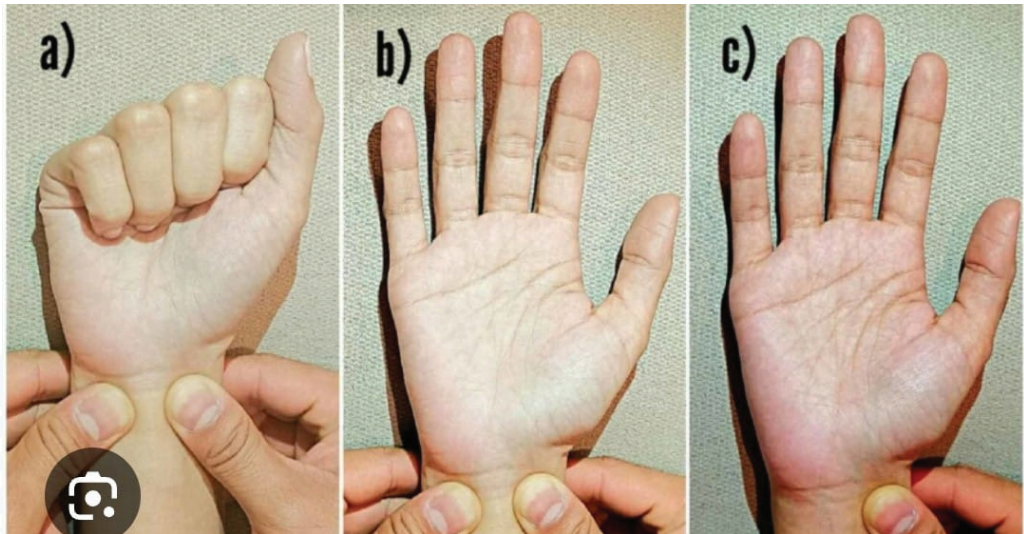
- ✦ Central venous pressure monitoring.
- ✦ Tip of the catheter is left between SVC and Right Atrium
- ✦ Monitors the right atrial pressure of the heart.

Transesophageal echocardiography



Active space

MIST

INTRA ARTERIAL MONITORING

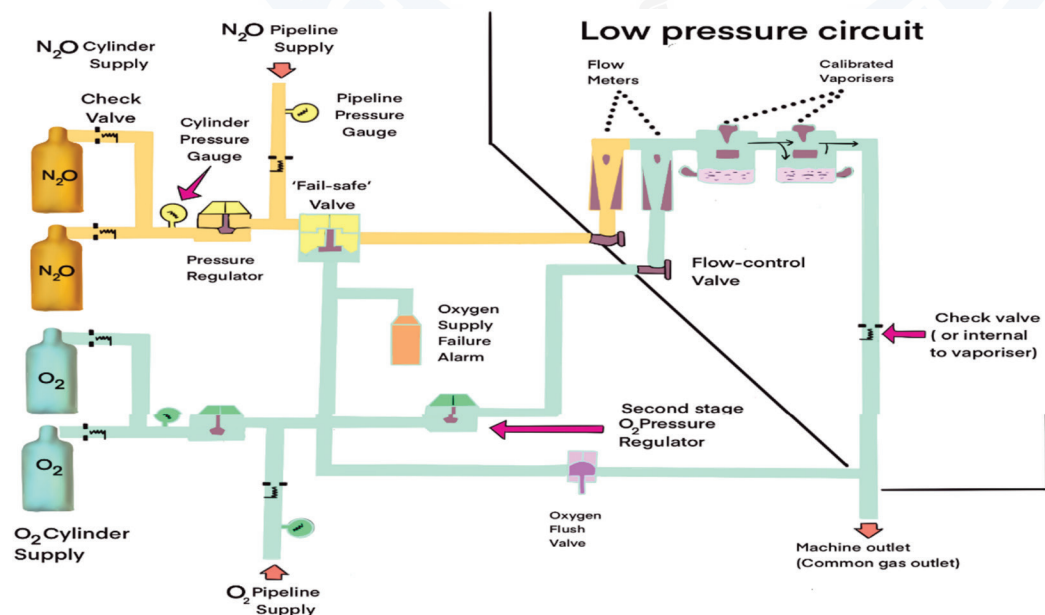
ANAESTHESIA MACHINE

Uses

- 1.
- 2.
- 3.
- 4.

• 1st Machine (1917)

- ➔ _____
- ➔ Boyle's apparatus



Pneumatic component

1. High pressure system

- a. Hanger yoke assembly
- b. Pressure gauge of cylinder (Bourdon's pressure gauge)
- c. Pressure regulator

2. Intermediate pressure system

- a. pipeline supply system
- b. Piping
- c. Pipeline pressure indicator
- d. DISS
- e. Emergency flush valve
- f. Oxygen failure indicators

Flow control valves 2nd pressure regulator

3. Low pressure system

- a. _____
- b. _____
- c. _____
- d. _____

Active space

Cylinders -

1. Size: A to HH

Body and shoulder are colour coded

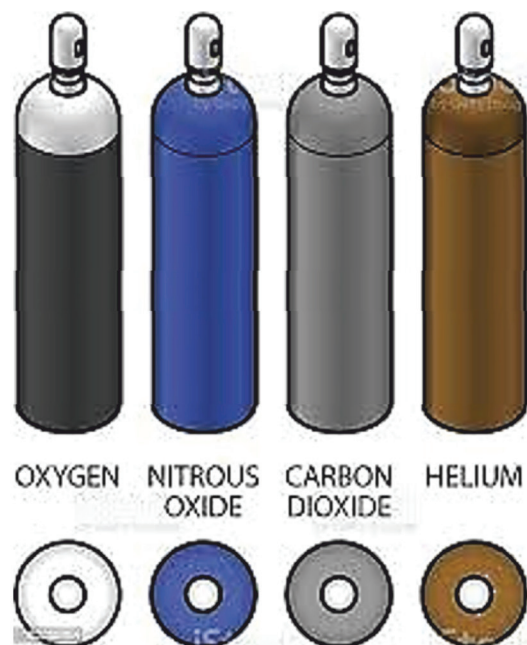
Label (most important)

Colour coding

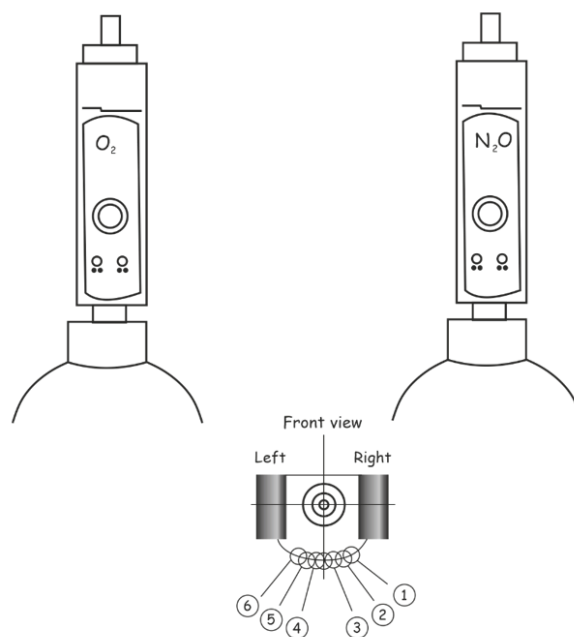
COLOR CODING		
GAS	SHOULDER	BODY
OXYGEN	WHITE	BLACK
NITROUS OXIDE	BLUE	BLUE
CYCLOPROPANE	ORANGE	ORANGE
CARBON DIOXIDE	GREY	GREY
AIR	WHITE	GREY
NITROGEN	BLACK	BLACK
ENTONOX	WHITE	BLUE

Active space

MIST



Pin index system: Prevents incorrect attachment of the cylinders



Pin index System	
Air	1, 5
O ₂	2, 5
N ₂ O	3, 5
CO ₂ (>7.5%)	1, 6
CO ₂ (<7.5%)	2, 6
Cyclopropane	3, 6
Entonox	7

Hanger Yoke Assembly system:

Has a centre port

Coming out from central port is nipple, which has a washer around it for a tight seal

2 pins below are unique for every particular gas Intermediate pressure system

1. Pipeline supply

Pipe color coding

White: O₂

Blue: N₂O

Black: Air

Yellow: Vacuum

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DISS: Diameter Index Safety System-This is a system which prevent incorrect attachment of pipeline to anaesthesia machine

2. _____

- a. Independent of electrical supply
- b.
- c.

3. O₂ fail safe device Activated when pressure of O₂ is below 30 psi, whistle starts blowing intermittently

4. Flow control valve: Controls flow of gas from machine -Follows colour coding

Low pressure system

Flowmeter

O₂ flowmeter is safe always to be kept _____ to other gases

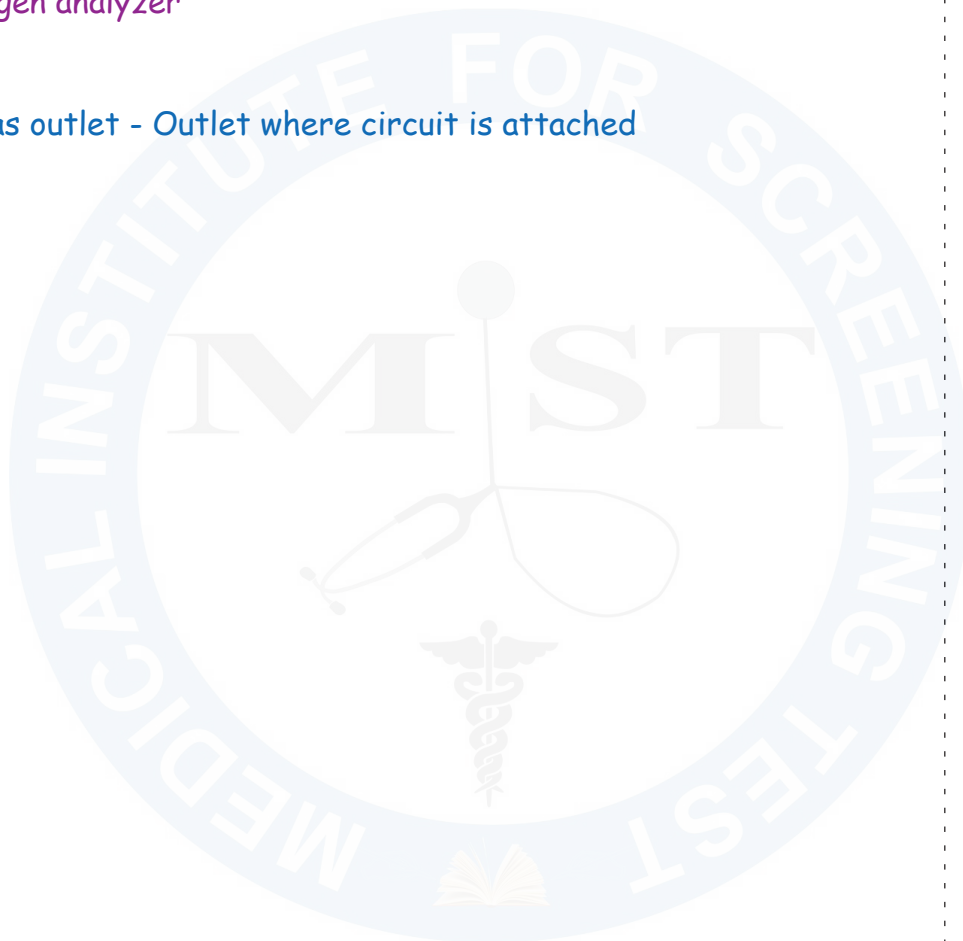
Active space

Vapouriser inhalational agent

Hypoxia guard

- a. Basal O_2 flow of anesthesia machine
- b. Link 25 systems, N_2O opens in fixed proportion with O_2
- c. Oxygen analyzer

Common gas outlet - Outlet where circuit is attached



Active space



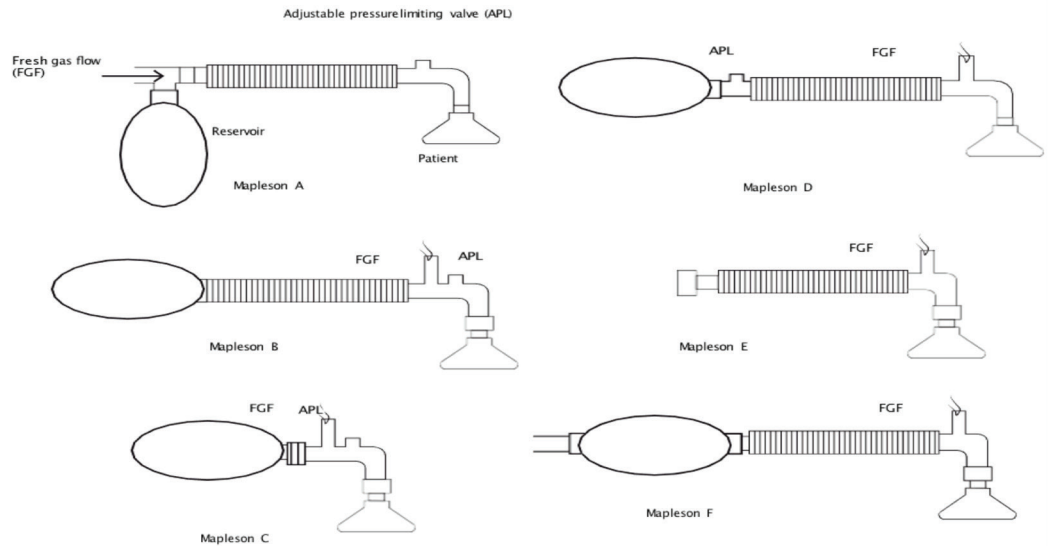
ANAESTHESIA CIRCUIT

- ✦ Anesthesia breathing circuit is attached to the common gas outlet
- ✦ It is a system which connects
- ✦ There are two types
 - ⇒ Mapleson system
 - ⇒ Circle system



Active space

MIST



Mapleson A

B

C

D

E

F

Active space

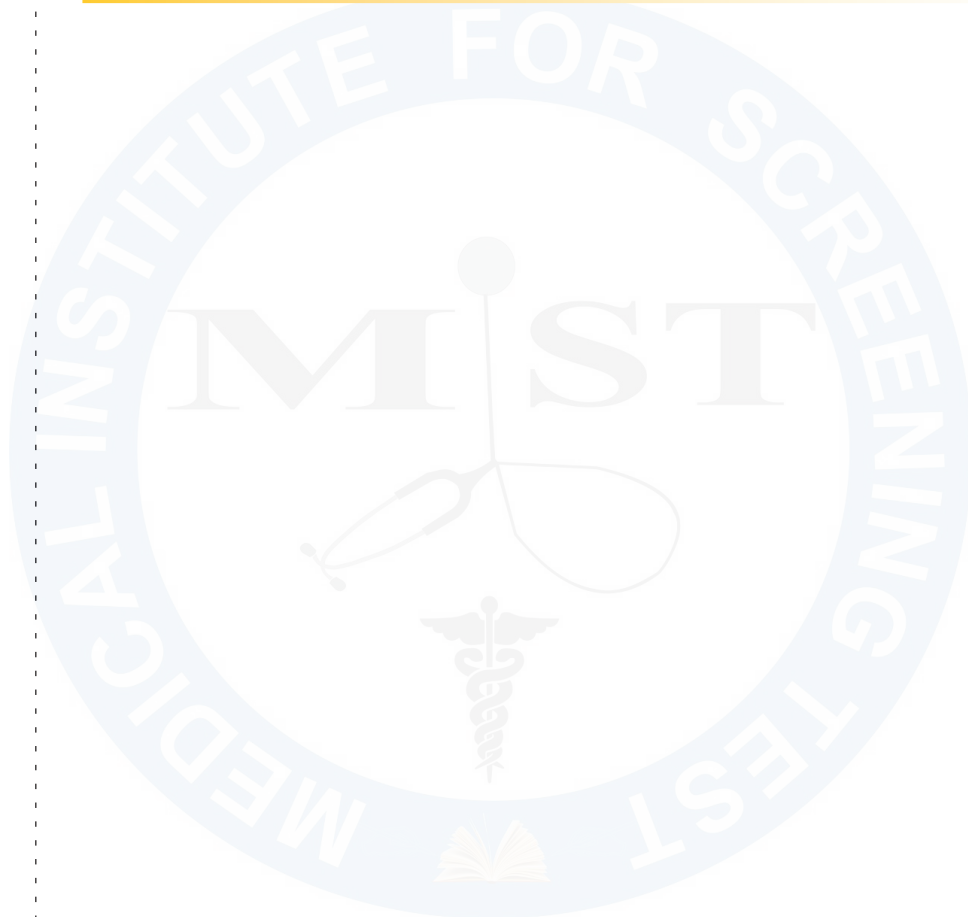
CARBONDIOXIDE ABSORBER

	Composition (%)
NaOH	4% (most effective)
KOH	1%
Ca(OH) ₂	Contains rest of the % (major constituent)
Water	11-18%
Silica	
Dye	



AIRWAY

AIRWAY

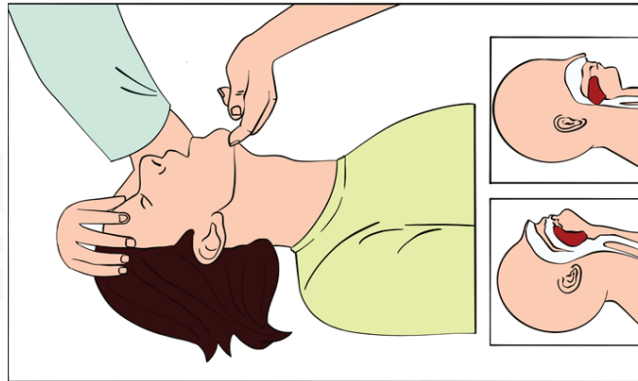


Active space

BASIC AIRWAY OPENING MANEUVERS

1. Head and chin Lift (HT/CL)

- One hand pushes the head backward and with other hand chin is lifted.



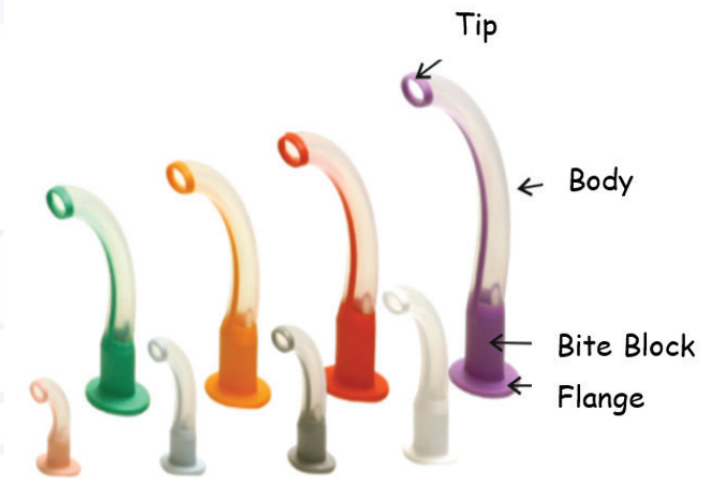
2. Jaw Thrust (JT)



- With both hand head is stabilized neutral position and Jaw is protruded forward

Simple Airway Adjunct

- ✦ Oropharyngeal Airway
- ✦ Most common is Guedel's airway.



- ✦ Guedel's airway is made up of rigid plastic
- ✦ Size is determined:
- ✦ Insertion:



NASOPHARYNGEAL AIRWAY

In a conscious patient with compromised airway, we use nasopharyngeal airway.

It is made up of soft silicon.

Size is determined:

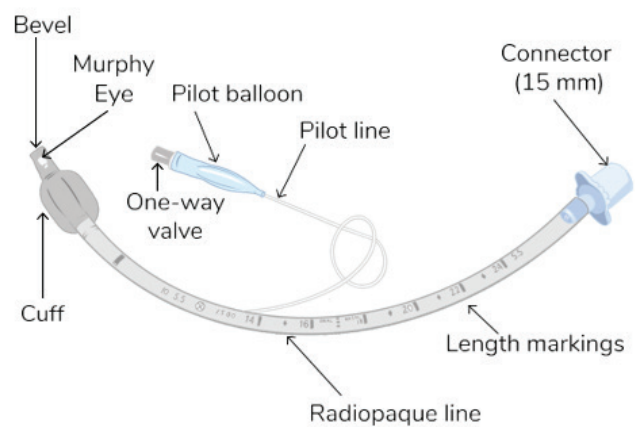
Insertion:



Endotracheal tube

ET tube through oral cavity: Needs Laryngoscope

Popularly used Laryngoscope is McIntosh (Curved Blade Laryngoscope)



Parts of ET tube

- ✦ Glottic opening: Bevelled to give a
- ✦ Murphy's eye:
- ✦ Cuff: Creates afor positive pressure ventilation.
- ✦ It also prevents any upper secretions of the upper airway to go into lower airways, thus preventing the aspiration.
- ✦ Black mark: Left at the glottic opening, for the correct positioning of the tube. ET tube is kept 3-4 cm above the carina.
- ✦ Universal connector: Connects to the circuit.

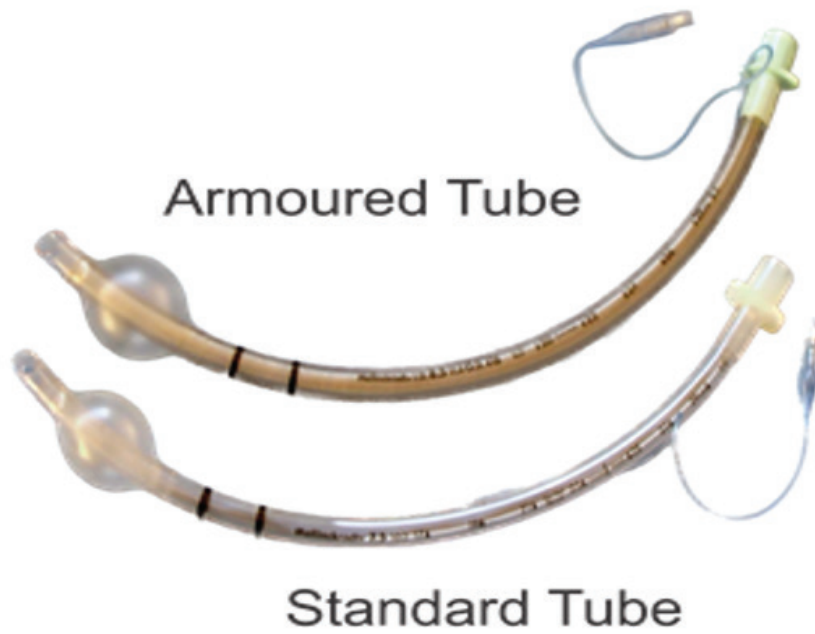
Steps of laryngoscopy

1. Positioning: Morning sniffing position/ barking dog position
Atlanto-occipital joint : extended.
Neck joint : flexed.
2. Hold the laryngoscope in the left hand.
3. Introduce the laryngoscope from right angle of mouth. Sweep the tongue from middle to left side.

4. Visualize Epiglottis and Arytenoid/ Vallecula.
5. Hinge the tip of Laryngoscope in Vallecula and lift it.
6. Inflate the cuff after introducing the tube.
7. Check for the right position either through auscultation or capnography.
8. Fix the tube

Confirmation of Right placement of ETT

Different types of ETT



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Management of Difficult Airway



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Alternate airway device

Classical LMA

✦ Designed by Dr Archie Brain

1.

2.

3.

4.

5.

6.

➔ Benefit -

➔ Disadvantage -

Second generation LMA



LMA classic



LMA unique



LMA Supreme



LMA Proseal



iLMA



i-gel

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- Benefit -
- Disadvantage -

MIST

SURGICAL AIRWAY

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General Anaesthesia

(Introduction)

Components:

STEPS:

1. Pre oxygenation

- 100% O₂ through Anaesthesia Mask and Circuit For (3-5 min)

2. Induction → Loss of consciousness

- Intravenous
- Inhalational

3. Maintenance → Maintenance of unconsciousness

- Intravenous
- Inhalational

4. Reversal

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INHALATIONAL ANAESTHETIC

Two groups

Potent inhalational agent

Carrier gas

New	Old	Carrier gas
Halothane	ether	N ₂ O Xenon
Methoxyflurane	chloroform	
Enflurane	cyclopropane	
Isoflurane		
Sevoflurane		
Desflurane		

Pharmacological properties

	Halothane	Isoflurane	Sevoflurane	Desflurane
Structure				
Boiling point				
Vapour pressure				

MAC



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Blood gas Partition Coefficient



Active space

MAC				
Blood: gas partition coefficient				
Stability				
Metabolism				

Pharmacodynamic properties

	Halothane	Isoflurane	Sevoflurane	Desflurane
CNS				
CVS				

RS				
Hepatic				
Renal				

- ♦ Malignant hyperthermia:
- ♦ Pharmacogenetic disease
- ♦ Triggering agents

1. _____
2. _____
3. _____

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Clinical features:

Increase metabolism

MIST

1. Increased CO_2
2. Increases BP
3. Increases HR
4. Arrhythmia
5. Increases core temperature
6. Increases muscle activity
7. HYPERKALEMIA
8. metabolic acidosis
9. myoglobinuria
10. Rhabdomyolysis
11. Renal failure

Management -

1. Stop triggering agent
2. IV Dantrolene



Classification

- ✦ They are divided into two groups
 - Opioids
 - Non-opioids - Thiopentone, Propofol, Etomidate and Ketamine

Opioids

- ✦ Only used for analgesic property in GA because of its side effect
- ✦ Can be used as sole anaesthetic only for major CVS compromised patient.
- ✦ It includes
 - Morphine
 - Fentanyl
 - Sufentanil
 - Alfentanil
 - Remifentanil
 - ⇒ BEST

- ❑ Unique metabolism, by non-specific esterase
- ❑ Ultrashort acting, 3-5 mins half-life (Use it as infusion)

Comparison of non-opioid anaesthetic agent

Pharmacokinetic properties

	Thiopentone	Propofol	Etomidate	Ketamine
Chemical nature				
Appearance				

Onset of action				
Duration of action				
Metabolism				

Thiopentone	Propofol	Etomidate	Ketamine
❑ CMRO ₂ (Cerebral metabolic directly proportional to requirement of O ₂) ❑ CBF ❑ ICP			Sympathomimetic ❑ CMRO ₂ ❑ CBF ❑ ICP ❑ EEG

Thiopentone Sodium and Propofol	etomidate	Ketamine
CVs unstable myocardial depressant	CVS stable	✦ CVS unstable: ↑ BP, ↑ HR ✦ Sympathomimetics because it inhibits catecholamines metabolism. ✦ But in absence of catecholamines, it becomes CVS depressant.

Effect on respiratory system

Thiopentone sodium	Propofol	Etomidate	Ketamine
Bronchospastic	All are Bronchodilator		

Contraindications

Thiopentone sodium	Propofol	Etomidate	Ketamine
✦ Shock ✦ Bronchial asthma ✦ Porphyria (as it is enzyme inducer)	✦ Shock	✦ Not used apart from CV compromised patient. ✦ Because it can cause adrenal suppression	✦ Ischemic heart disease ✦ Hypertension ✦ Epilepsy ✦ Elective Neuro surgery ✦ Glaucoma ✦ Critically ill patient

Uses

Active space

Side effects

Thiopentone sodium	Propofol	Etomidate	Ketamine

Active space



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NEUROMUSCULAR BLOCKERS

Acts on NMJ (Neuromuscular Junction)

Depolarizing	Non - Depolarizing
Succinylcholine	Rest All
Persistent / prolonged depolarization of Nc (Nicotinic cholinergic receptor).	Acts by competitive antagonism.
Fasciculation is seen. Post-operative myalgia may be seen. Not reversed by Neostigmine.	No fasciculations seen. No myalgia seen. Reversal by Neostigmine.

Depolarizing	Non-Depolarizing		
	Aminosteroids	Benzylisoquinolines	Oximes
succinylcholine	Pancuronium	D-tubocurarine	Gantacurium
	Pipecuronium	Doxacurium	
	Vecuronium	Atracurium	
	Rocuronium	Cis-atracurium	
	Rapacuronium	Mivacurium	



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SUCCINYLCHOLINE

- Onset of action: 30-40 sec
- Duration of action: 6-8 min

IV SUCCINYLCHOLINE

- 5-15% reach in NMJ (6-8 min)

SUCCINYLCHOLINE APNEA

- Single normal dose of succinyl choline produced prolonged apnoea
- Deficiency of pseudocholinesterase
- Atypical pseudicholinesterase
- Rx: Mechanical ventilation

Phase 1 and 2 block

Side effects:

- Fasciculation
- Postoperative myalgia
- Intraoperative pressure
- Intracranial pressure
- Intra gastric pressure
- Hyperkalemia /Exaggerated Hyperkalemia
- Bradycardia

Use

1. RSI



Active space

MIST

Reversed of G A



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REGIONAL ANAESTHESIA

LOCAL ANAESTHETICS

- It acts on sodium channel → closes the sodium channel → stop propagation of nerve impulse

Additives

- LA + vasoconstrictor(adrenaline)

- Advantage :

• _____ • _____ • _____ • _____

- Disadvantage :

o _____ o _____ o _____

Sensitivity of different nerve fibre to local anaesthetics

Amino Ester	Amino Amide
Short Acting • Cocaine • Procaine • Chloroprocaine	Intermediate Acting • Lignocaine • Mepivacaine • Prilocaine
Long Acting • Benzocaine • Tetracaine	Long Acting • Bupivacaine • Levobupivacaine • Ropivacaine • Dibucaine
• All except Cocaine are metabolized by pseudo cholinesterase. • Cocaine is metabolized by hepatic metabolism.	• All are metabolized by hepatic metabolism.



Active space

MIST

Complications of Local Anaesthetics

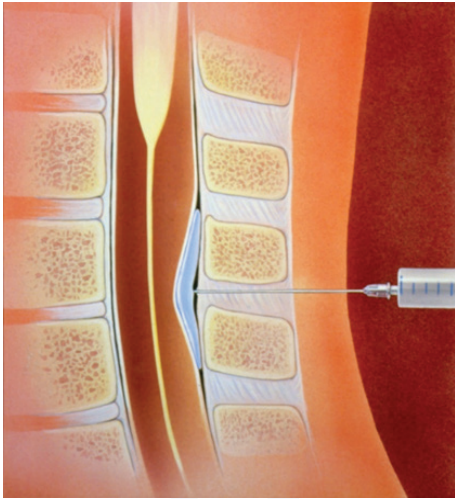
1. LAST- Local anaesthetic systemic toxicity-
CNS toxicity -

CVS toxicity -

Management -

3. Methemoglobinaemia -

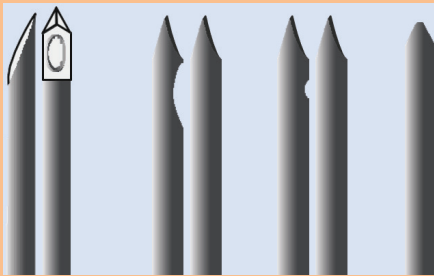
CENTRAL NEURAXIAL BLOCKADE



SPINAL AND EPIDURAL ANESTHESIA

S.A (Drug between pia & arachnoid mater in subarachnoid space)	E.A (Drug outside dura mater)
Drug goes into CSF and very large number of nerves are blocked.	Catheter can be placed to obtain longer duration anesthesia
Onset: Fast.	Onset: Delayed
Small amount of drug needed.	Large amount of drug needed. Per spinal dermatome required 1.5-2.0 ml/drug
Dense block.	Patchy block.
NO systemic toxicity.	More systemic toxicity.

Active space

Single shot technique.	Not a single shot technique, Duration can be prolonged
Only Intra-Operative use.	Intra operative anesthesia + Post operative analgesia
Uniform block.	Segmental block.
More hemodynamic.	Less hemodynamic.
• Imbalance: Less Stability. • Easy and less incidence of failure.	• Imbalance: More stability. • More expertise needed and more chances of failure.
Spinal needle • Dura cutting: Quincke needle. • Dura splitting: Sprotte needle and Whitacre needle. 	Epidural needle • Tuohy's needle (markings present on the needle).

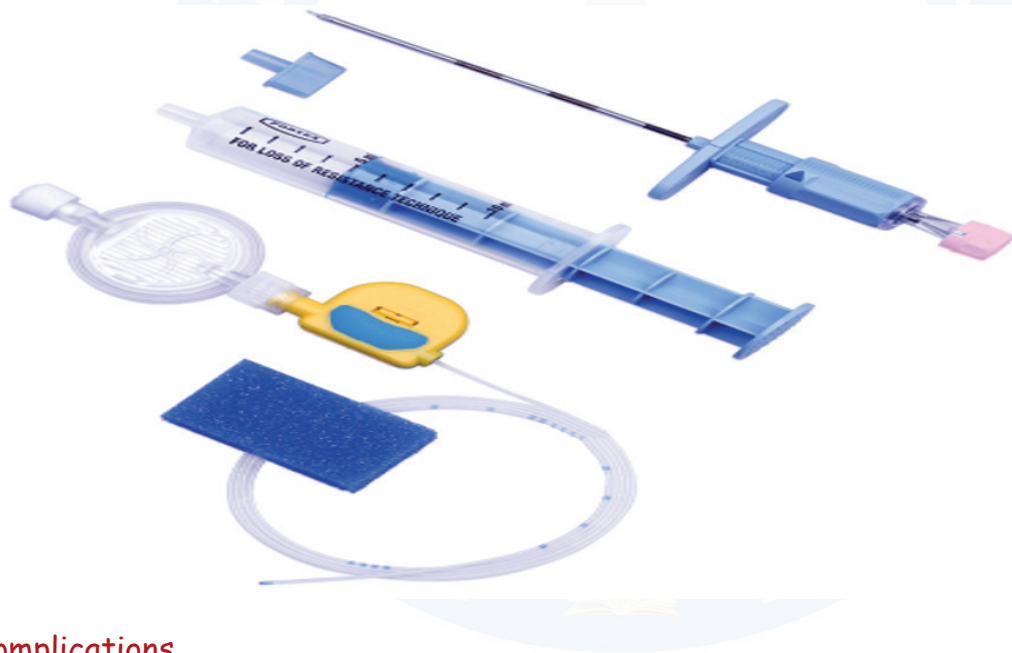
Spinal Anesthesia Procedures

1. Skin > subcutaneous fascia > supraspinous ligament > interspinous ligament > ligamentum flavum > dura mater > arachnoid mater > subarachnoid space.
2. Remove the stellate.
3. Check for free flow of CSF.
4. Administer the drug.

5. Remove the needle.

Epidural Anesthesia Procedures

- ✦ Skin > subcutaneous fascia > supraspinous ligament > interspinous ligament > ligamentum flavum > epidural space.
- ✦ Loss of resistance is felt through the syringe attached to the needle.
- ✦ Epidural does not have a visual elicit like spinal anesthesia drug administration, therefore chances of failure is more.



Complications

1. Most Common: Hypotension
 - ✦ Treatment: Phenyl epinephrine.
2. Bradycardia
 - Treatment: Atropine.
3. Respiratory depression.

4. Urinary Retention: Most common post operative complication.
5. Total Spinal Anesthesia/High Spinal Anesthesia
 - a. All 31 spinal nerve roots are blocked.
 - b. Hypotension + Bradycardia + Respiratory Distress.
6. Systemic toxicity to L.A - Only with Epidural Anesthesia.
7. Infection
8. Vasovagal shock.
9. Post - Dural puncture headache - now a days rare, because of the use of dura splitting needle.

Contraindication of central neuraxial blockade

- 1.
- 2.
- 3.
- 4.
- 5.
- 6.

OXYGEN THERAPY

Low flow device: Low flow and variable FiO₂

- a. Nasal cannula.
- b. Simple face mask.
- c. Face mask with reservoir bag.

High flow device: High flow and fixed FiO₂

- d. Venturi mask.
- e. Nebulizers.
- f. High flow nasal cannula.
- g. Self-inflatable bag valve mask (Ambu-bag).

Nasal cannula



Simple face mask



Face tent



Oxymizer



Venturi mask



Non-rebreather



High flow nasal cannula



Bipap



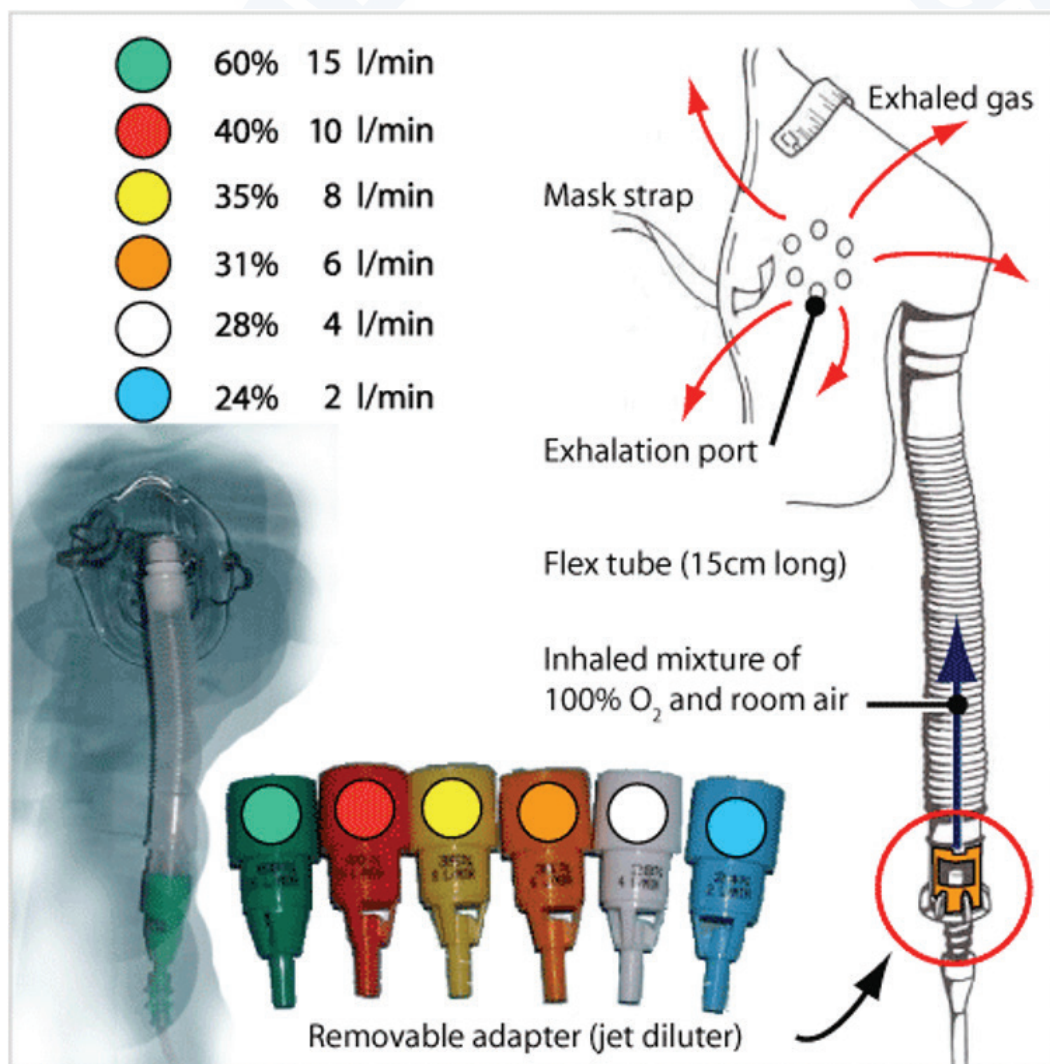
Nasal cannula - Flow rate -
FiO₂ -
Indications -

Simple face mask - Flow rate -
FiO₂ -
Indications

Nonrebreathing mask - Flow rate -
 FiO₂ -
 Indications

High flow devices -

Venturi mask



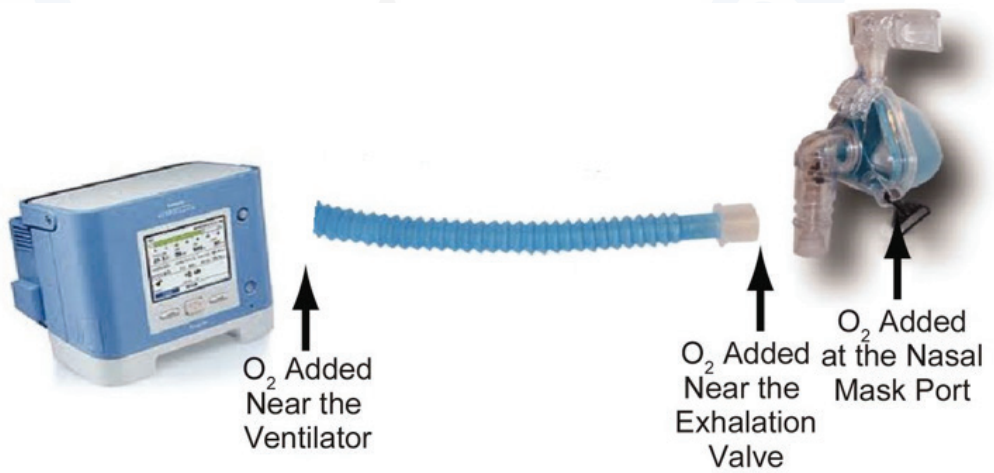
Active space

MIST

HFNC-



NIV



MIST

PERIOPERATIVE FLUID

Difference between crystalloid and colloid

INTRAVENOUS CANNULA

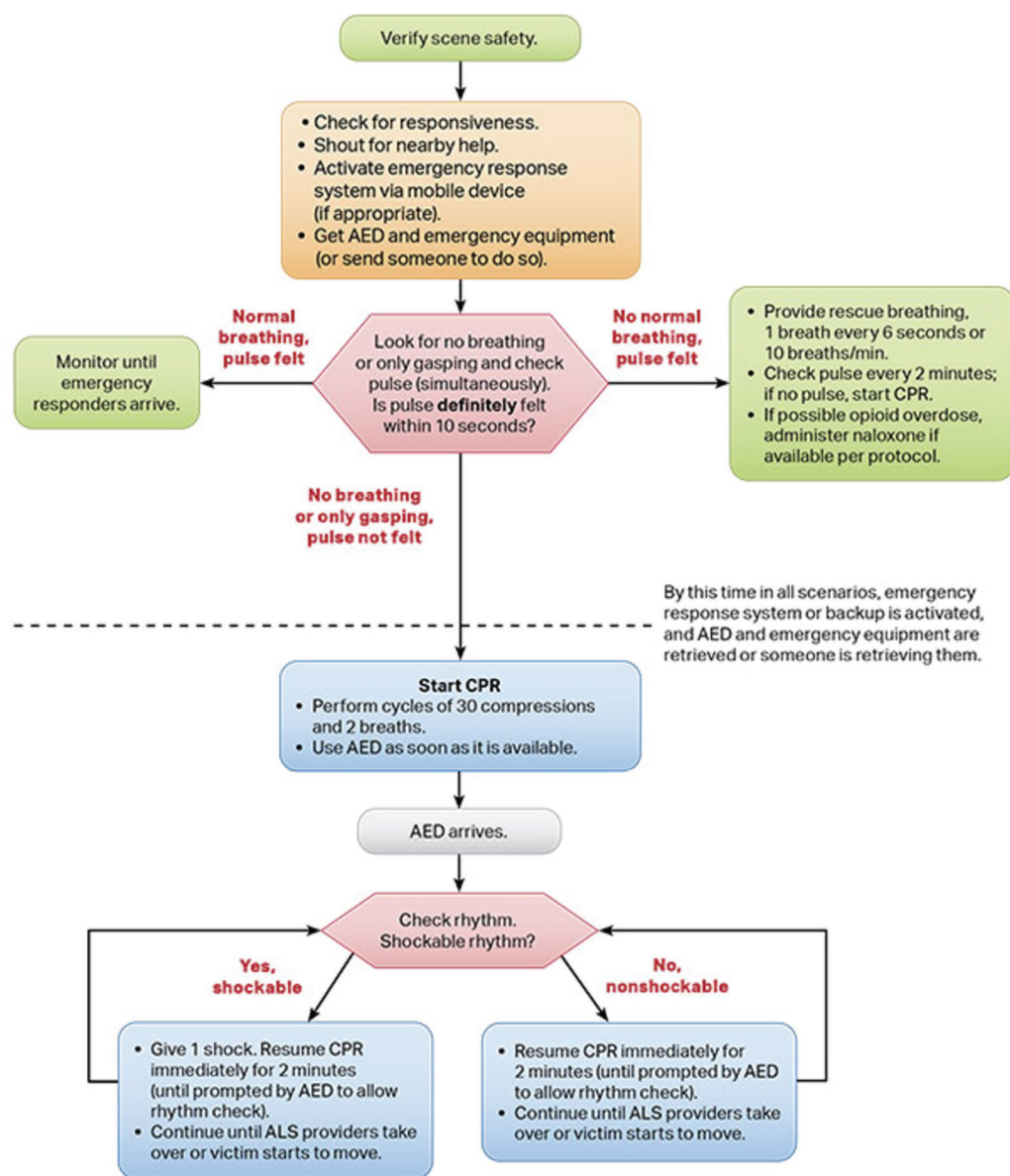
Gauge	Colour Code	Ext. Dia. mm	Length mm	Flow Rate ml/min
14G	Orange	2.1	45	240
16G	Grey	1.8	45	180
18G	Green	1.3	32/45	90
20G	Pink	1.1	32	60
22G	Blue	0.9	25	36
24G	Yellow	0.7	19	20
26G	Violet	0.6	19	13

Active space

APPROACH TO UNDIFFERENTIATED EMERGENCY PATIENT

Basic life support

Adult Basic Life Support Algorithm for Healthcare Providers



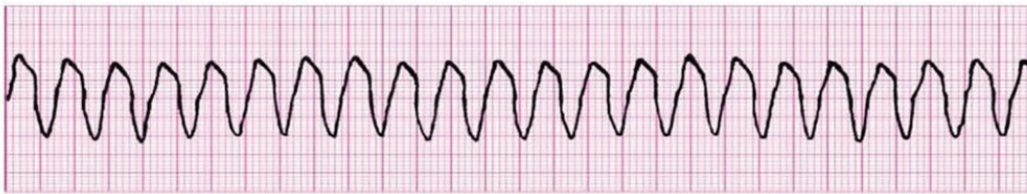
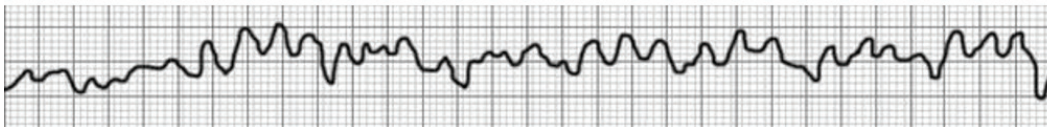
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Advanced cardiac life support

Management depends on type of rhythm

Two types of rhythm

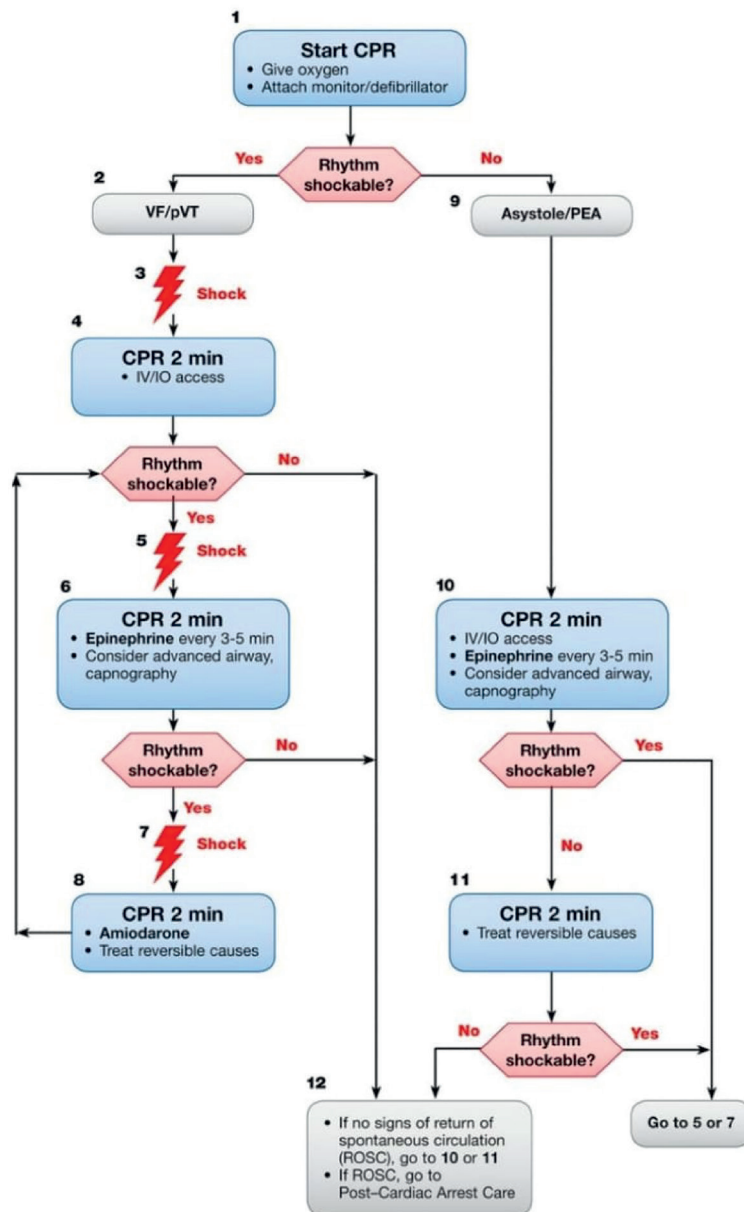
1. Shockable -



2. Non shockable

Active space

Adult Cardiac Arrest Algorithm—2015 Update



CPR Quality

- Push hard (at least 2 inches [5 cm]) and fast (100-120/min) and allow complete chest recoil.
- Minimize interruptions in compressions.
- Avoid excessive ventilation.
- Rotate compressor every 2 minutes, or sooner if fatigued.
- If no advanced airway, 30:2 compression-ventilation ratio.
- Quantitative waveform capnography
 - If PETCO_2 <10 mm Hg, attempt to improve CPR quality.
- Intra-arterial pressure
 - If relaxation phase (diastolic) pressure <20 mm Hg, attempt to improve CPR quality.

Shock Energy for Defibrillation

- **Biphasic:** Manufacturer recommendation (eg, initial dose of 120-200 J); if unknown, use maximum available. Second and subsequent doses should be equivalent, and higher doses may be considered.
- **Monophasic:** 360 J

Drug Therapy

- **Epinephrine IV/IO dose:** 1 mg every 3-5 minutes
- **Amiodarone IV/IO dose:** First dose: 300 mg bolus. Second dose: 150 mg.

Advanced Airway

- Endotracheal intubation or supraglottic advanced airway
- Waveform capnography or capnometry to confirm and monitor ET tube placement
- Once advanced airway in place, give 1 breath every 6 seconds (10 breaths/min) with continuous chest compressions

Return of Spontaneous Circulation (ROSC)

- Pulse and blood pressure
- Abrupt sustained increase in PETCO_2 (typically >40 mm Hg)
- Spontaneous arterial pressure waves with intra-arterial monitoring

Reversible Causes

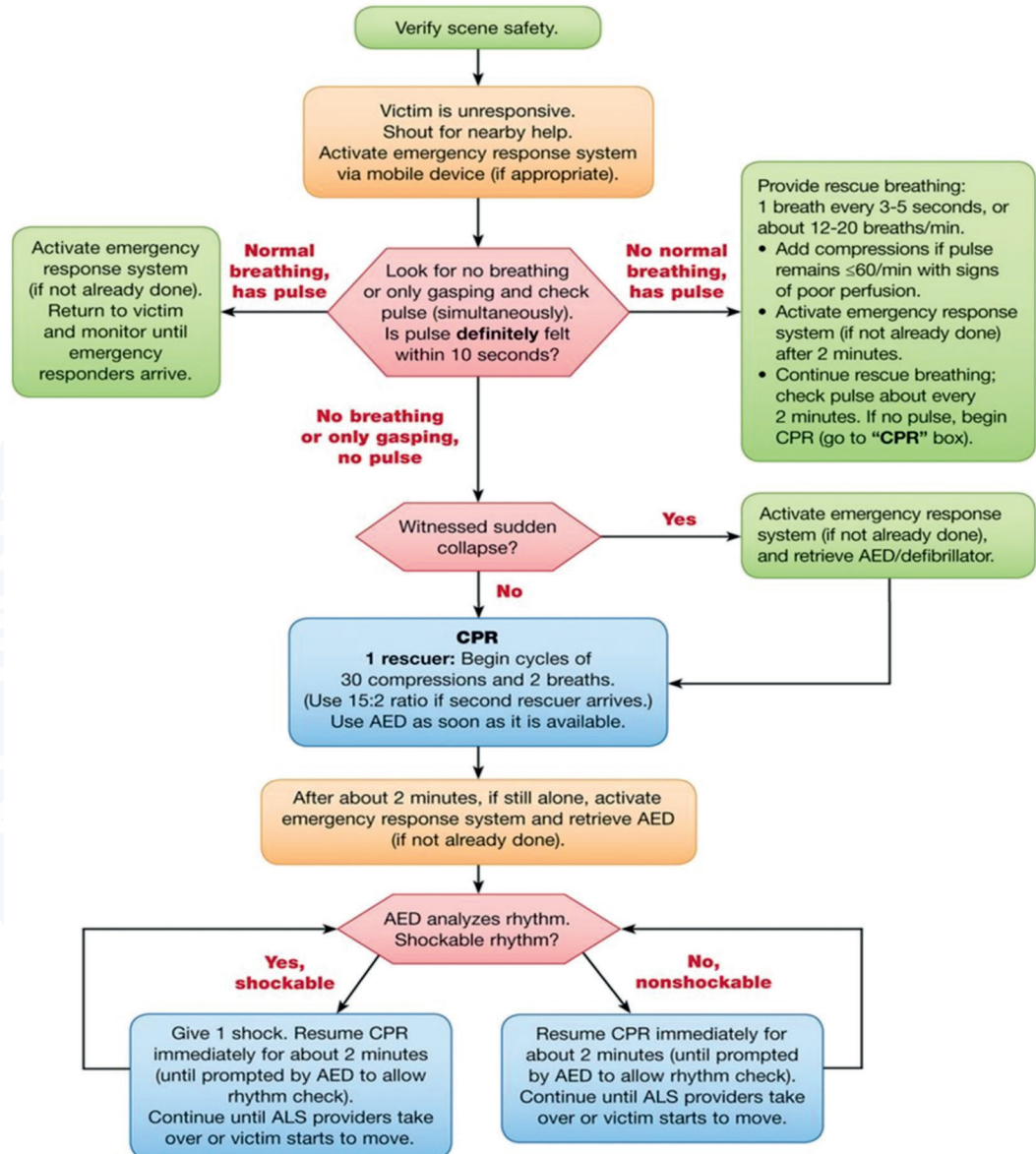
- Hypovolemia
- Hypoxia
- Hydrogen ion (acidosis)
- Hypo-/hyperkalemia
- Hypothermia
- Tension pneumothorax
- Tamponade, cardiac
- Toxins
- Thrombosis, pulmonary
- Thrombosis, coronary



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BLS Healthcare Provider Pediatric Cardiac Arrest Algorithm for the Single Rescuer—2015 Update



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